

WELLINGTON, New Zealand

WMO: 934360

Lat: 41.3311S

Lon: 174.8060E

Elev: 7

StdP: 101.24

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 00488

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.1	3.9	-0.2	3.7	8.3	0.8	4.0	8.4	18.9	8.8	17.3	8.9	5.8	10	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.0	23.9	18.0	22.8	17.4	21.6	17.0	19.4	22.1	18.6	21.3	17.9	20.6	8.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.1	20.5	17.3	12.4	20.0	16.9	12.0	19.7	55.1	22.1	52.7	21.2	50.6	20.6	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.3	13.1	DB	1.7	26.5	1.1	1.3	0.9	27.4	0.3	28.2	-0.3	28.9	-1.1	29.8	(4)
			WB	0.9	20.8	1.0	0.9	0.2	21.4	-0.4	22.0	-1.0	22.5	-1.7	23.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	17.3	17.5	16.3	14.4	12.8	10.7	9.8	10.3	11.6	12.8	14.3	16.4
	DBStd	3.55	2.89	3.21	2.86	2.43	2.29	2.18	2.02	1.93	2.06	2.01	2.07	2.17
	HDD10.0	94	3	4	3	2	4	18	29	20	8	3	1	0
	HDD18.3	1759	49	43	72	119	171	229	264	249	202	171	123	66
	CDD10.0	1430	229	213	198	133	91	38	23	29	56	90	128	199
	CDD18.3	54	18	19	9	1	0	0	0	0	0	0	1	7
	CDH23.3	57	24	19	6	0	0	0	0	0	0	0	1	7
	CDH26.7	1	1	0	0	0	0	0	0	0	0	0	0	1
Wind	WSAvg	6.9	7.2	6.7	6.7	6.4	6.6	6.6	6.5	6.5	7.1	7.5	7.6	7.3
Precipitation	PrecAvg	1208	73	65	82	94	117	135	134	128	102	108	85	84
	PrecMax	1607	136	347	158	184	197	216	265	203	193	256	195	191
	PrecMin	974	10	5	24	28	16	38	15	28	21	34	16	32
	PrecStd	161	35	66	38	40	47	44	59	45	41	56	48	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	25.1	24.1	20.9	19.0	16.6	15.7	15.9	17.2	19.2	21.2	24.1
		MCWB	19.5	18.5	18.0	17.0	15.6	13.8	12.7	12.2	13.0	14.3	15.9	17.8
	2%	DB	23.9	23.9	22.2	19.8	18.0	15.9	14.8	14.9	16.2	17.9	19.8	22.1
		MCWB	17.9	18.2	17.0	16.1	15.2	13.4	12.0	11.8	12.8	13.5	15.0	16.7
	5%	DB	22.2	22.8	21.1	18.9	17.1	15.0	14.0	14.1	15.2	16.9	18.2	20.9
		MCWB	17.3	17.6	16.9	15.6	14.4	12.6	11.3	11.2	12.4	13.3	14.2	16.4
	10%	DB	21.1	21.2	20.1	18.0	16.2	14.1	13.1	13.2	14.9	16.0	17.5	19.9
		MCWB	16.9	17.3	16.4	15.1	13.6	11.7	10.6	10.6	12.2	12.9	13.8	15.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.6	19.4	18.2	16.8	14.6	13.8	13.4	14.6	15.6	17.2	19.2
		MCDB	23.7	22.9	22.0	19.8	18.1	16.0	15.1	14.8	16.3	17.5	19.7	21.9
	2%	WB	19.3	19.5	18.3	17.0	15.6	13.6	12.6	12.5	13.8	14.8	16.0	18.1
		MCDB	22.2	22.0	21.1	18.8	17.2	15.3	14.2	14.2	15.5	16.8	18.4	20.6
	5%	WB	18.5	18.7	17.6	16.1	14.9	12.9	11.6	11.7	13.0	14.0	15.2	17.4
		MCDB	21.3	21.3	20.2	18.2	16.7	14.6	13.4	13.6	15.0	16.1	17.5	19.8
	10%	WB	17.6	17.9	16.9	15.4	14.0	12.2	11.0	11.1	12.4	13.2	14.5	16.7
		MCDB	20.3	20.8	19.5	17.5	16.0	14.0	12.9	13.1	14.5	15.5	16.8	19.1
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.0	5.9	5.3	4.9	4.8	4.9	5.0	5.0	5.4	5.7	5.6
		MCDBR	7.6	7.6	7.1	5.9	5.2	4.9	5.1	5.6	5.5	6.1	6.8	7.0
		MCWBR	4.0	3.8	3.9	3.9	3.9	4.0	4.0	4.0	3.9	4.1	4.2	3.9
	5% WB	MCDBR	6.6	6.0	5.7	4.8	4.4	4.2	4.3	4.6	4.9	5.1	5.6	5.7
		MCWBR	4.1	3.9	3.9	3.8	3.8	3.9	3.9	3.8	4.0	4.0	4.0	3.8
Clear-Sky Solar Irradiance	taub	0.348	0.335	0.324	0.312	0.296	0.295	0.294	0.305	0.327	0.343	0.334	0.344	
	taud	2.547	2.606	2.628	2.650	2.662	2.634	2.635	2.616	2.536	2.506	2.556	2.563	
	Ebn at Noon	980	967	936	882	831	795	823	874	915	948	988	989	
	Edh at Noon	106	96	85	72	60	57	61	73	92	104	105	106	
All-Sky Solar Radiation	RadAvg	6.67	5.88	4.60	3.06	2.07	1.54	1.78	2.60	3.80	5.11	6.33	6.58	
	RadStd	0.43	0.50	0.28	0.22	0.12	0.09	0.13	0.15	0.23	0.35	0.49	0.45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	+0.53	N/A	N/A	N/A	N/A	-23	N/A	N/A	N/A
	Regional (7 neighbors)	N/A	N/A	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page